

Yiran Hu

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SUMMARY

Ph.D. researcher with 5 years of research experience in software engineering and AI, focusing on LLM coding agents. Experienced in agent framework and skill design, large-scale evaluation, cost-efficiency optimization, training data synthesis, and LLM fine-tuning. Published at ISSRE, ACL, and ICRA (**Best Paper Award Finalist**).

EDUCATION

Purdue University, West Lafayette, IN Aug 2023 – May 2028 (Expected)
Ph.D. in Computer Science · Advisor: [Prof. Bharat Bhargava](#)
Tianjin University, Tianjin, China Sep 2019 – Jun 2023
B.S. in Computer Science and Technology · GPA: 89.45/100 (3.73/4.0) · Advisor: [Prof. Junjie Chen](#)

RESEARCH EXPERIENCE

Purdue University – Graduate Research Assistant Aug 2023 – Present

Analyzing and Optimizing Cost Efficiency of Coding Agents [Under Review, first author] Jan 2026 – Aug 2026

- Built an efficiency analysis pipeline and identified three cost-inefficient behaviors (subsumed retrieval, similar script generation, test re-execution) wasting up to 22.75% of task cost in 1,200 Claude Code and mini-SWE-agent trajectories.
- Evaluated structure-aware retrieval (CodeGraph) and agent-synthesized skills as cost optimizations across 10K+ agent runs in 32–96 parallel containers on SWE-bench Verified and Pro, finding inconsistent savings.
- Designed developer-written agent skills to improve agent behavioral efficiency, cutting task cost by up to 41.73% and the targeted behaviors by up to 88.91% while maintaining Pass@1.

TENET: Test-Driven Agent for Repository-Level Code Generation [ISSRE 2026, first author] Feb 2025 – Sep 2025

- Designed TENET, a test-driven agentic framework for repository-level code generation, whose test harness selects a concise test subset by caller diversity and invocation proximity, beating the strongest baseline by 9.49 pp in Pass@1.
- Built a retrieval (Tree-sitter AST) and debugging (pdb) toolset for navigating large codebases and diagnosing failures; removing it drops Pass@1 by 14.89 pp.
- Designed a reflection-based refinement workflow that analyzes test failures to iteratively revise code; removing it drops Pass@1 by 9.24 pp.

REPOCOD: Repository-Level Code Generation Benchmark [ACL 2025] Oct 2024 – Mar 2025

- Curated 980 code-generation tasks from 11 popular Python repositories with developer-written tests to benchmark LLMs on realistic, repository-level coding tasks.
- Built repository-specific Docker environments for reproducible, execution-based evaluation that double as verifiable-reward environments for agent training.
- Evaluated commercial LLMs with RAG (including BM25) to compare different retrieval approaches, showing none of the 10 LLMs exceeded 30% Pass@1.

SELP: Safe and Efficient LLM Planners for Robot Agents [ICRA 2025] Sep 2023 – Sep 2024

- Implemented an equivalence-voting mechanism to make LLM-generated Linear Temporal Logic (LTL) specifications more reliable, raising translation accuracy from 88.4% to 98.0%.
- Synthesized drone-navigation and tabletop-manipulation datasets via grammar-based LTL generation and LLM paraphrasing to train LLM planners on complex, safety-constrained tasks.
- Contributed to supervised fine-tuning (SFT) of Llama-family translators and planners on the synthesized data; SELP improves safety rate by up to 20.4% over prior LLM planners.

Software Analysis and Intelligence Lab, Tianjin University – Undergraduate Researcher Nov 2021 – May 2023

- Adapted LEMON, a deep-learning library testing tool, from TensorFlow to MindSpore, uncovering 8 bugs.
- Applied differential testing to the LLVM and TVM compilers to detect compiler bugs.
- Optimized ResNet-50 with TVM graph- and operator-level tuning, reducing inference latency by 14.4 ms.

INDUSTRY EXPERIENCE

Zhongke Wenge Science and Technology Co., Ltd., Beijing, China

May 2022 – Aug 2022

Algorithm Engineer Intern, Research & Development Center

- Diagnosed PaddleOCR failure modes on COCO-Text and SVT to locate accuracy bottlenecks.
- Integrated SAST detection with SVTR recognition trained on ICDAR2015 to handle distorted text, improving detection accuracy by 27.8%.
- Added a recognition module for detectors previously incompatible with recognition, improving accuracy by 15.4%.

PUBLICATIONS

- [P1] **Analyzing and Mitigating Cost-Inefficient Behaviors in Coding Agents** **Under Review**
Yiran Hu, Nan Jiang, Shanchao Liang, Anik Dey, Yi Wu, Lin Tan. [PDF]
- [C1] **TENET: One Step Toward Test-Driven Development for Repository-Level Code Generation** **ISSRE 2026**
Yiran Hu, Nan Jiang, Shanchao Liang, Yi Wu, Lin Tan. [PDF]
- [C2] **Can Language Models Replace Programmers for Coding? REPOCOD Says ‘Not Yet’** **ACL 2025**
Shanchao Liang, Nan Jiang, Yiran Hu, Lin Tan. [PDF]
- [C3] **SELP: Generating Safe and Efficient Task Plans for Robot Agents with LLMs** **ICRA 2025**
Yi Wu, Zikang Xiong, Yiran Hu, Shreyash S. Iyengar, Nan Jiang, Aniket Bera, Lin Tan, Suresh Jagannathan. [PDF]
Best Paper Award Finalist
- [C4] **LocRDF: An Ontology-Aware Key-Value Store for Massive RDF Data** **WISA 2022**
Yunxiao Zhang, Jinghan Li, Xueyang Liu, Rong Cheng, Yiran Hu, Xin Wang. [PDF]

COURSE PROJECTS

- Test-Time Compute Scaling for LLM Math Reasoning** **Spring 2026**
- Compared Self-Consistency and PRM-weighted Best-of-N on MATH-500 (Llama-3.2-1B/3B, Llama-3.1-8B PRM, $n=1-64$) to study when a reward model beats majority voting.
 - Found that correct answers exist in 73%–81% of candidate pools but the majority voting mechanism in Self-Consistency often misses them; PRM-weighted selection beats it by 5.4 pp on the 1B model.
- Vulnerability Detection with Explanation-Aware Fine-Tuning and Preference Optimization** **Spring 2025**
- Fine-tuned Qwen2-0.5B-Instruct on PrimeVul (7.6K paired samples) to detect and explain vulnerabilities, comparing four strategies: label-only SFT, SFT with CVE explanations, SFT with synthetic explanations, and GRPO.
 - Built a data augmentation pipeline in which a teacher model (Qwen2-1.5B-Instruct) writes explanations for benign code, sampling 4 candidates per function and keeping the one with the highest average token log-probability.
 - Trained with SFT and GRPO on 4 A5000 GPUs with DeepSpeed; synthetic explanations raised paired-test F1 from 0% to 20.4% while accuracy stayed near 50%, exposing the limits of sub-1B models.

TEACHING

Teaching Assistant, Purdue University

Aug 2023 – Present

CS 176 Data Engineering in Python · CS 251/253 Data Structures & Algorithms · CS 571 Artificial Intelligence

HONORS & AWARDS

- Best Paper Award Finalist**, IEEE International Conference on Robotics and Automation (ICRA) 2025
- Outstanding Graduate, Tianjin University 2023
- First Prize, Challenge Cup National College Student Sci-Tech Competition (Tianjin University round) 2022
- First Class Scholarship, Tianjin University 2021

SKILLS

Languages: Python, C++, Bash

LLM Agents: Claude Code, Codex, Google ADK, SWE-agent, OpenHands, MCP, Anthropic/OpenAI APIs

Agent System Design: Agent Orchestration, Toolset Design, Prompt Engineering, RAG

LLM & ML: PyTorch, Hugging Face, Fine-Tuning (SFT), Reinforcement Learning (GRPO), Synthetic Data Generation

Evaluation: Benchmark Design, Static & Dynamic Analysis, Agent Trajectory Analysis, Cost Efficiency Analysis

Infrastructure: Docker, Linux, Git, pytest, Tree-sitter, DeepSpeed